

MOTIONCORE · PROBLEM STATEMENT

MotionCore problem statement

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-02	Amish Chadha	Brushed and dual-motor host questions decided (MTC-DEC-001, 2026-10-02)
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). Input range 20 to 60 V (CellGuard 16S covered); MTC-DDR-001 items recorded as decided
0.3	2026-09-25	Amish Chadha	TRL 3. Budget scope (motor costed to hosts), SwapCell and CellGuard links, first host and input range adopted for TRL 3 pending Amish's review (MTC-DDR-001); CargoMule pack corrected to 12S LiFePO4
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users, host projects, context, constraints, prior work, open questions)
0.1	2026-09-25	Amish Chadha	Initial scaffold

MotionCore problem statement

Each vehicle or machine concept in the lab re-specifies the same drive train and safety logic, and small teams rarely have time to get the emergency stop, speed limit and brake interlock right. MotionCore is a shared controller and safety module, with a named reference hub motor and one documented interface, so that each host design inherits a reviewed safety chain instead of inventing its own.

■ The problem

Five existing lab designs drive a motor from a lithium pack: CargoMule (a 36 V, 250 W hub motor in a bicycle trailer), PalletPilot (two 24 V hub motors on a pallet jack), StepClimber (a 24 V worm gearmotor on a stair-climbing hand truck), SunSpoke (a 48 V, 250 W hub motor on a roadster bicycle) and DustRunner (12.8 V gearmotors on a panel-cleaning robot). Each one specifies its own controller, fuse, emergency stop, brake cut-off and speed limit, in its own words, with its own gaps. Three problems follow:

1. **Repeated effort.** The same parts (a brushless controller, a contactor, a fuse, a mushroom-head stop button, brake switches) are selected, wired and documented five times, and a fix found in one design does not reach the others.
2. **Uneven safety logic.** The safety function is the part most often left vague. Typical gaps in small builds are an emergency stop that only asks the firmware to stop, a speed limit held in the same controller that could fail, no interlock between the brake and the drive, and a restart that happens by itself when power returns. Machinery standards such as [ISO 13850](#) (emergency stop) and [ISO 13849-1](#) (safety-related control parts) describe how to avoid these, but they are written for industrial teams and are costly to read.
3. **Rising exposure.** Light electric vehicles and small machines are spreading fast. The IEA reports almost 10 million electric two-wheelers sold worldwide in 2025, about 14 % of all two-wheeler sales ([IEA Global EV Outlook 2026](#)). In the United States, the Consumer Product Safety Commission estimates about 698,500 emergency department visits linked to micromobility products from 2017 through 2024 and records 533 deaths, 310 of them involving e-bikes ([CPSC, 2017 to 2024 report](#)). These figures cover all causes, most of them crashes, so they show the scale of exposure rather than a failure rate of drive electronics.

■ Users and context

User	Need	Context
Lab project lead (host designer)	Drop in a drive and safety chain with known limits and a documented interface; spend design time on what is new about the host	Designing a vehicle or machine at TRL 2 to 4
Builder of a host prototype	Wire, configure and test the drive in a garage with hand tools and a multimeter, and know when it is safe to power up	Home or makerspace workshop, first power-up on a bench
Operator of a host machine	Stop the machine at once from a clearly marked button; never have it start by itself	Walk-behind machines (pallet jack, stair climber), cargo trailers, bicycles
People near the machine	Not be struck by a machine that runs away, over-speeds or restarts	Warehouses, sidewalks, stairwells, farms
Maintainer or repairer	Diagnose a fault from a fault log and swap a part without special tools or closed software	Local repair shop or the lab

Host projects (first-order envelope)

Table 1 lists the drive needs stated in the host READMEs. MotionCore is sized to cover the first four; DustRunner's 12.8 V system falls outside the 20 to 60 V input range.

Table 1. Drive needs of existing lab host designs.

Host	Pack	Motor	Stop and brake need
SunSpoke	48 V SwapCell (about 46.8 V nominal, 54.6 V full)	250 W geared front hub	Brake cut-off on both levers; pedelec speed cutoff
CargoMule	12S LiFePO4 (38.4 V nominal), 384 Wh	250 W geared hub, 20 in wheel	Drive cut when the drawbar goes into compression; overrun brakes
PalletPilot	25.6 V LiFePO4, 20 Ah	Two 24 V hub motors with spring-applied brakes	Two hardwired e-stops, safety relay, walking-pace limit
StepClimber	24 V LiFePO4, 10 Ah	24 V worm gearmotor with spring-applied brake	Stop on tilt; brake holds on stairs
DustRunner	12.8 V LiFePO4	Two small drive gearmotors	Out of the 20 to 60 V range

Operating environment

- **Supply:** LiFePO4 or lithium-ion packs from 8S LiFePO4 (about 25.6 V nominal, 29.2 V full) to 13S lithium-ion (54.6 V full) and 16S LiFePO4 (58.4 V full), including the portfolio's SwapCell pack and CellGuard-managed packs.
- **Power:** 250 W continuous reference, up to 500 W continuous on 24 V hosts; short peaks to about 750 W.
- **Climate:** 0 to 40 °C ambient in the working range, rain and dust on outdoor hosts, vibration from rough roads and stairs.
- **Speeds:** walking pace (about 1.5 m/s, 5.4 km/h) for walk-behind machines; up to 25 km/h for pedal-assist hosts.

■ Constraints

- Garage-buildable prototype, \$300 or less for the MotionCore kit (module and devices); the reference motor is costed to each host (MTC-DDR-001 item 1).
- Off-the-shelf, openly documented parts: a VESC-class controller ([VESC project](#)), a DC contactor, standard fuses and industrial pushbuttons.
- The emergency stop must work without any firmware: a hardwired loop that opens the main contactor.
- One connector set shared by all hosts; keyed so that the safety loop and the command lines cannot be swapped.
- Compatible with SwapCell interface v0.3 (the supervisor acts as the vehicle host) and the CellGuard BMS where a host uses them (decided by Amish, 2026-09-25, MTC-DDR-001 items 7 and 8).
- Research prototype only. MotionCore does not make a host compliant with any machinery, vehicle or pedelec regulation; each host remains responsible for its own risk assessment.

■ Out of scope

- Battery packs and their BMS (SwapCell and CellGuard cover these).
- Host-specific mechanics: wheels, frames, brakes, gearboxes and guards.

- Traction above about 1 kW or systems above 60 V DC.
- Certification to ISO 13849, IEC 61508 or vehicle type approval.
- Autonomy, navigation and obstacle sensing (hosts such as PalletPilot own those layers and feed a stop request into the MotionCore safety loop).

■ Prior work

- **VESC.** An open brushless motor controller started by Benjamin Vedder, with open hardware designs and firmware for DC, BLDC and FOC control ([VESC project](#); [firmware](#), [GPL-3.0](#)). It already has current, temperature and speed (ERPM) limits, CAN bus and a PC tool, and many low-cost variants are sold. It has no separate safety channel.
- **ODrive and SimpleFOC.** Open motor control for robotics ([ODrive](#); [SimpleFOC](#)). Good for position control; like VESC, they leave the safety chain to the integrator.
- **Machinery safety practice.** [ISO 13850:2015](#) sets the principles for emergency stop design; [IEC 60204-1](#) defines stop categories 0, 1 and 2; [ISO 13849-1:2023](#) sets the architecture and performance levels for safety-related control parts. MotionCore borrows their structure (two channels, fail-safe contactor, no automatic restart) without claiming conformity.
- **Pedelec rules.** In the European Union, pedal cycles with assistance up to 250 W that cuts off at 25 km/h are excluded from type approval under [Regulation \(EU\) No 168/2013](#); EN 15194 sets the product standard. This sets the default speed cutoff for pedal-assist hosts.
- **Commercial safety relays and drives.** Industrial drives with Safe Torque Off inputs and dual-channel safety relays solve the problem well, at a cost and size that suit factories, not garage prototypes.

■ Open questions

On 2026-09-25 Amish decided to go with the recommendations on first host, SwapCell heartbeat, stop category and input range (MTC-DDR-001, MTC-DDR-002): CargoMule is the first adopter, subject to its project's agreement; the supervisor sends the SwapCell heartbeat; category 0 is the default with category 1 as a per-host option; and the range is 20 to 60 V, which covers CellGuard's 16S LiFePO4 packs, so DustRunner stays out of scope. Still open:

- Do the host leads agree to swap their current drive sections for MotionCore, starting with CargoMule?
- Can StepClimber's brushed worm gearmotor run from the VESC DC motor mode, or does it need a different power stage? Decided by Amish, 2026-10-02 (MTC-DEC-001): DC motor mode, unless StepClimber's motor needs more current than the controller is rated for.
- Should a dual-motor host such as PalletPilot use two controllers on one supervisor, or two modules? Decided by Amish, 2026-10-02 (MTC-DEC-001): two controllers on one supervisor, with one safety loop and one contactor feeding both.

■ User research and co-design

MotionCore's first users are the lab's own host designers, so co-design starts inside the lab.

- [] Review Table 1 with each host project and confirm voltage, power, speed and stop needs
- [] Walk through the safety loop with a builder who has not seen it and record where it is unclear
- [] Ask at least one external maker or repair shop to review the interface and fault chart

- [] Revise requirements (MTC-REQ-001) from findings before freezing the interface